Role of Artificial Intelligence in Commerce and Management

The following material can be used for your research paper. It includes currently available statistics, India-focused facts, applications, benefits, challenges, and a ready-made research framework. According to the latest available reports, AI is no longer limited to the technology sector; it is rapidly transforming marketing, finance, human resources, supply chain management, retail, and strategic management.

Suggested Research Title

"Role of Artificial Intelligence in Commerce and Management: Opportunities, Challenges and Future Prospects with Special Reference to India"

Abstract

Artificial Intelligence (AI) is making modern commerce and management more data-driven, automated, and customer-centric. It is being used in market forecasting, customer relationship management, fraud detection, inventory management, financial analysis, recruitment, performance evaluation, and strategic decision-making.

AI adoption is increasing rapidly among large enterprises in India. According to IBM, 59% of large enterprises in India were actively using AI, while an additional 27% of organizations were exploring its adoption.

According to the World Economic Forum's *Future of Jobs Report 2025*, 86% of employers expect AI and information-processing technologies to transform their businesses by 2030. The report also estimates that 170 million new jobs could be created by 2030, while 92 million jobs could be displaced, resulting in a net increase of 78 million jobs.

This research paper examines the applications, benefits, risks, and future implications of AI in commerce and management. The study concludes that AI is not a complete replacement for human decision-making but rather a technology that supports human intelligence and managerial judgement.

Keywords: Artificial Intelligence, Commerce, Management, Machine Learning, Generative AI, Marketing, Finance, Human Resources, Business Analytics, India.

1. Introduction

Artificial Intelligence is a technology that enables machines to learn, reason, understand language, make predictions, and perform decision-making tasks. Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, and Generative AI are its major components.

In commerce, AI is used to understand customers, increase sales, determine pricing strategies, prevent fraud, and improve supply chain operations. In management, it is being used to make planning, organizing, staffing, directing, and controlling functions more efficient.

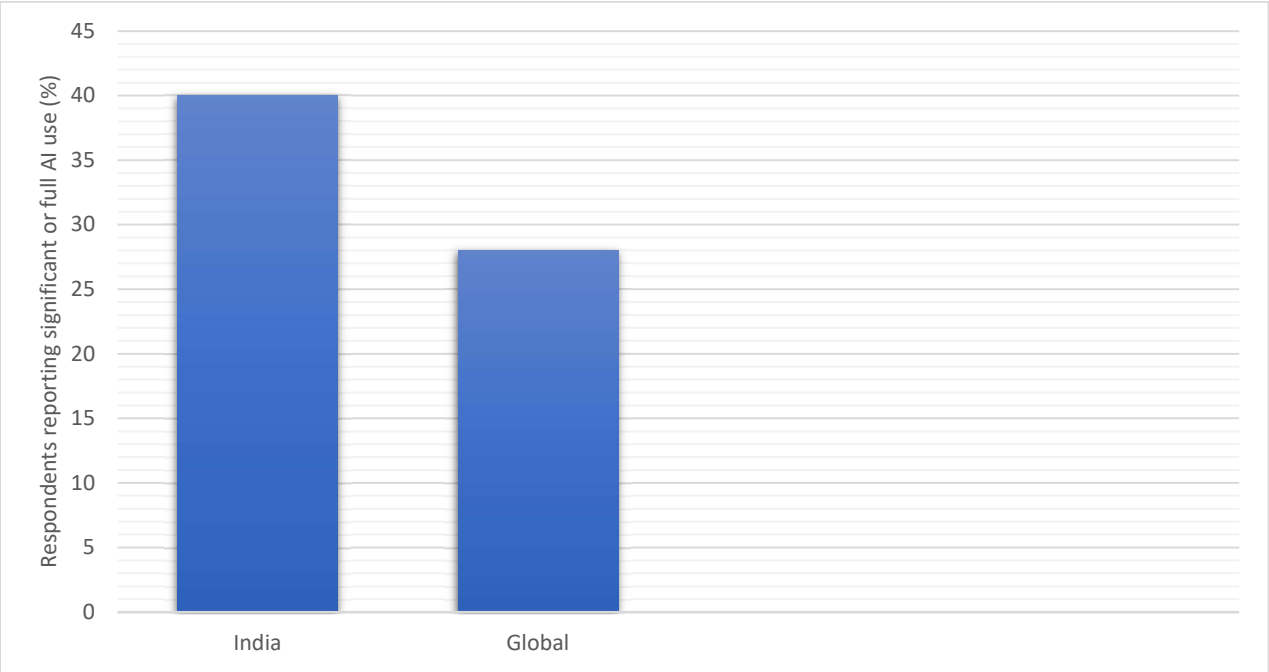
To promote AI adoption in India, the Government of India approved the IndiaAI Mission in March 2024 with a budget outlay of ₹10,371.92 crore. According to the Press Information Bureau (PIB), Ministry of Electronics and Information Technology, as of 25 March 2026, more than 38,000 GPUs had been onboarded through the IndiaAI Compute Portal under the Mission. These GPUs are being made available to Indian startups and academic institutions at affordable rates to support AI research, innovation, and development. This initiative aims to expand access to advanced computing infrastructure and promote the growth of India's AI ecosystem.

2. Current Facts and Figures

Indicator	Current Figure	Research Relevance
Active AI deployment among large Indian enterprises	59%	Shows the current status of business AI adoption in India
Large Indian enterprises exploring AI	27%	Indicates future adoption potential
IndiaAI Mission budget	₹10,371.92 crore	Demonstrates government support
Employers expecting AI to transform their businesses by 2030	86%	Indicates global business transformation
Potential new jobs by 2030	170 million	Highlights employment opportunities
Potential displaced jobs by 2030	92 million	Highlights employment disruption risks
Net employment change due to AI	78 million increase	Shows the overall labour-market impact
Indian respondents reporting significant or full AI use	Approximately 40%	Shows India's comparative adoption position
Global respondents reporting significant or full AI use	28%	Provides a global comparison
Indian Generative AI market size in 2025	Approximately USD 2.23 billion	Indicates market growth
Projected Indian Generative AI market size by 2030	Approximately USD 41.51 billion	Represents an estimated CAGR of 51.9%

According to Deloitte India's 2026 AI report, approximately 40% of Indian respondents reported significant or full AI usage, compared with the global average of 28%. The same report stated that 56% of organizations had implemented AI at scale in strategy and operations, 62% in product development, and 55% in marketing and sales.

The Indian Generative AI market was valued at approximately USD 2.23 billion in 2025 and is projected to reach approximately USD 41.51 billion by 2030, representing an estimated CAGR of 51.9%. These market projections depend on the methodology used by the source and should therefore be presented in research papers as estimated market forecasts rather than confirmed future figures.



3. AI in Commerce

3.1 Marketing and Advertising

AI analysis customer data, browsing history, and purchasing behaviour to support personalized marketing.

Major applications:

- Customer segmentation.
- Personalized advertisements.
- Product recommendations.
- Email and social media campaign automation.
- Customer sentiment analysis.
- Demand and sales forecasting.
- Dynamic pricing.

For example, e-commerce platforms can recommend products based on customers' previous purchases and search behaviour. This can improve customer experience, conversion rates, and repeat purchases.

According to McKinsey's 2025 report, the highest reported revenue benefits from AI were found in marketing and sales, strategy and corporate finance, and product and service development.

3.2 Retail and E-Commerce

AI is useful in retail for the following activities:

- Inventory optimization.
- Stock-out prediction.
- Warehouse automation.
- Chatbots and virtual shopping assistants.
- Demand forecasting.
- Identification of fraudulent transactions.

- Last-mile delivery optimization.
- Customer churn prediction.

AI helps retailers estimate which products will experience increased demand, when demand will increase, and in which geographical areas. This helps reduce both overstocking and understocking.

3.3 Banking and Financial Services

The use of AI in commerce and finance is increasing rapidly.

Major applications:

- Credit scoring.
- Loan approval assistance.
- Fraud detection.
- Anti-money laundering monitoring.
- Algorithmic trading.
- Risk assessment.
- Insurance claim processing.
- Personal financial advisory services.

AI can analysis large volumes of financial transactions in real time. If a transaction differs from a customer's usual spending behaviour, the system can flag it as a suspicious transaction.

3.4 Accounting and Auditing

AI can automate routine and repetitive accounting tasks.

- Invoice processing.
- Expense categorization.
- Bank reconciliation.
- Tax calculation assistance.
- Duplicate payment detection.
- Financial statement analysis.
- Audit risk identification.
- Cash flow forecasting.

This saves time for accountants and allows them to focus more on higher-value activities such as compliance, financial interpretation, and strategic financial planning.

3.5 Supply Chain and Logistics

AI makes supply chain management more predictive and efficient.

- Demand forecasting.
- Supplier selection.
- Route optimization.
- Delivery-time prediction.
- Warehouse robotics.
- Quality inspection.
- Procurement analytics.
- Disruption prediction.

If a supplier is likely to delay a delivery, AI can analysis historical data and external factors to alert management in advance.

4. AI in Management

4.1 Strategic Management

AI provides managers with real-time insights to support business decisions.

Applications of AI:

- Competitor analysis.
- Market trend identification.
- Scenario planning.
- Business forecasting.
- New market selection.
- Mergers and acquisitions analysis.
- Strategic risk assessment.

However, human judgement remains essential in final strategic decisions because business decisions also involve factors such as ethics, organizational culture, and social impact.

4.2 Human Resource Management

AI is used in Human Resource Management (HRM), from recruitment to employee retention.

- Resume screening.
- Candidate matching.
- Interview scheduling.
- Employee onboarding.
- Training recommendations.
- Performance analytics.
- Employee attrition prediction.
- Employee engagement analysis.
- Payroll and attendance automation.

AI can help HR managers predict which employees may be at risk of leaving an organization. However, biased recruitment algorithms can result in unfair treatment of qualified candidates.

4.3 Operations Management

Operations managers can use AI to improve productivity, quality, and operational efficiency.

- Predictive maintenance.
- Production scheduling.
- Defect detection.
- Resource allocation.
- Process automation.
- Energy optimization.
- Quality control.

In manufacturing, predictive maintenance can provide warnings before machinery breaks down. This can reduce downtime and maintenance costs.

4.4 Decision-Making

Traditional management decisions were often based on limited information and historical reports. AI helps managers make decisions using large datasets, dashboards, and predictive models.

AI-based decision-support systems can help answer questions such as:

- Which product is more profitable?
- Which customers are likely to stop purchasing from the business?
- Which market should the organization expand into?
- Which employees require additional training?

- Which suppliers present greater risks?

However, AI-generated outputs should not be accepted blindly because incorrect, incomplete, or biased data can lead to incorrect decisions.

5. Major Benefits of Artificial Intelligence

5.1 Higher Productivity

AI completes routine tasks more quickly. This allows employees to focus more on analysis, creativity, and customer service instead of manual data entry.

According to Deloitte India's 2026 survey, approximately 97% of respondents expected AI to increase productivity.

5.2 Cost Reduction

Automation can reduce processing time, paperwork, and operational expenses. It can have a particularly significant impact on departments such as finance, customer support, logistics, and human resources.

5.3 Better Customer Experience

AI chatbots can provide 24/7 customer support. Recommendation engines display relevant products to customers, while personalized communication can improve customer satisfaction.

5.4 Faster Decision-Making

Through real-time data analysis, AI provides managers with information more quickly. This enables businesses to respond more effectively to changing market conditions.

5.5 Improved Forecasting

AI can analysis past sales, customer behaviour, seasonality, and market trends to forecast demand, sales, cash flow, and employee turnover.

5.6 Innovation

Generative AI can support product design, content creation, software development, market research, and business model innovation.

According to McKinsey's 2025 findings, 64% of respondents stated that AI was enabling innovation in their organizations.

6. Challenges and Risks of Artificial Intelligence

6.1 Data Privacy

AI systems depend on large amounts of customer and employee data. If data is collected, stored, or shared improperly, it can create serious privacy risks.

According to Deloitte India's report, more than 70% of respondents considered data security and privacy to be high or very high concerns.

6.2 Algorithmic Bias

If training data is biased, AI can cause discrimination in recruitment, lending, pricing, or insurance decisions.

Examples:

- Giving lower preference to candidates of a particular gender during recruitment.

- Considering customers from certain geographical areas to have higher credit risk.
- Providing limited chatbot support to certain languages and communities.

6.3 Lack of Transparency

Many AI models operate like black boxes. Managers may find it difficult to understand why an AI system has produced a particular decision or recommendation.

6.4 Cybersecurity Threats

AI systems may be vulnerable to hacking, data poisoning, prompt injection, and misinformation. Therefore, organizations should establish access controls, encryption, monitoring, and audit systems.

6.5 Job Displacement

AI may affect certain repetitive and rule-based jobs. According to the World Economic Forum, 92 million jobs could be displaced by 2030, although 170 million new jobs are expected to be created.

6.6 Hallucination and Incorrect Output

Generative AI sometimes produces answers that appear fluent and believable but are factually incorrect. Therefore, human verification is essential before using AI outputs for financial, legal, medical, and strategic decisions.

6.7 High Implementation Cost

AI adoption requires organizations to invest in:

- Data infrastructure.
- Cloud computing.
- Skilled employees.
- Cybersecurity.
- Software integration.
- Training and change management.

According to Deloitte, the major investment priorities of Indian organizations for AI scalability included security and compliance controls at 68%, data storage and management at 61%, and scalable infrastructure and computing capacity at 54%.

7. Role of Managers in the AI Era

The role of managers will not disappear in the AI era; instead, it will evolve. Rather than focusing only on controlling operations, managers will need to integrate technology, people, and ethics. Modern managers should:

- Develop a basic understanding of AI tools.
- Learn data-driven decision-making.
- Reskill and upskill employees.
- Verify AI-generated results.
- Implement data privacy and ethical standards.
- Maintain human creativity and empathy.
- Measure the return on investment (ROI) of AI projects.

According to the World Economic Forum, 77% of employers plan to upskill their workers by 2030, while approximately 40% of employers expect to reduce their workforce in roles that can be automated by AI.

8. India-Specific Relevance

AI is particularly important for commerce and management in India because the country has:

- A large digital consumer base.
- Widespread use of UPI and digital payments.
- Rapidly growing e-commerce.
- Multilingual customers.
- A large Micro, Small and Medium Enterprises (MSME) sector.
- Availability of IT and software talent.
- Indian businesses operating in global markets.

AI can make customer support, rural banking, agricultural commerce, healthcare commerce, and government services more accessible through Indian languages.

However, cost, data quality, digital skills, and infrastructure remain major challenges for small businesses.

9. Conceptual Framework

The following conceptual framework can be used in your research paper:

Independent Variable:

AI Adoption

AI Dimensions:

- Machine Learning.
- Generative AI.
- Predictive Analytics.
- Natural Language Processing.
- Robotic Process Automation.

Mediating Variables:

- Data quality.
- Employee digital skills.
- Management support.
- IT infrastructure.
- Ethical governance.

Dependent Variables:

- Business performance.
- Operational efficiency.
- Customer satisfaction.
- Decision-making quality.
- Innovation capability.
- Employee productivity.

Conceptual Framework Flow



10. Research Objectives

The following objectives can be included in the research paper:

1. To study the role of Artificial Intelligence in commerce and management.

2. To analysis AI applications in marketing, finance, human resources, and operations.
3. To study the impact of AI adoption on business performance.
4. To understand the productivity and cost benefits resulting from AI.
5. To identify the major challenges and ethical risks associated with AI adoption.
6. To examine the future prospects of AI in India.
7. To identify the essential AI skills required by managers and employees.

11. Research Questions

1. In which areas of commerce and management is AI being used most extensively?
2. Does AI improve business efficiency and productivity?
3. How does AI influence customer satisfaction and decision-making?
4. What are the major organizational and technological barriers to AI adoption?
5. Will AI replace existing jobs or create new employment opportunities?
6. How prepared are Indian businesses for AI adoption?
7. What is the role of data privacy and ethical governance in AI adoption?

12. Hypotheses

The following hypotheses can be used in the research paper:

H1: Artificial Intelligence adoption has a significant positive effect on organizational performance.

H2: AI-based customer analytics has a significant positive effect on customer satisfaction.

H3: AI adoption significantly improves operational efficiency.

H4: Employee digital skills strengthen the positive relationship between AI adoption and business performance.

H5: Data privacy concerns negatively affect AI adoption.

H6: Top-management support positively influences the successful implementation of AI.

13. Suggested Research Methodology

13.1 Research Design

A descriptive and analytical research design would be suitable for this topic.

13.2 Data Collection

Primary Data:

- Structured questionnaire.
- Managers.
- Employees.
- Business owners.
- Commerce students.
- Banking and retail professionals.

Secondary Data:

- Government reports.
- World Economic Forum reports.
- Deloitte reports.
- IBM reports.
- Annual reports of companies.
- Research journals.
- Industry reports.

13.3 Sample Size

For practical research purposes, the following sample sizes can be considered:

- 100–200 respondents: Suitable for undergraduate research papers.
- 250–400 respondents: Suitable for a strong dissertation-level study.
- More than 400 respondents: Suitable for a more reliable comparative study.

For sample collection in Bhopal, the study can include banks, retail stores, e-commerce users, MBA students, accountants, and small-business owners.

13.4 Questionnaire Scale

A five-point Likert scale can be used:

1. Strongly Disagree.
2. Disagree.
3. Neutral.
4. Agree.
5. Strongly Agree.

Sample Statements:

- AI increases the productivity of our organization.
- AI improves customer service.
- AI makes decision-making more accurate.
- AI automates routine management tasks.
- Our employees have sufficient skills to use AI.
- Data privacy concerns create barriers to AI adoption.
- Senior management supports AI adoption.
- AI improves business innovation.

13.5 Statistical Tools

The following statistical tools can be used:

- Percentage analysis.
- Mean and standard deviation.
- Cronbach's alpha.
- Correlation analysis.
- Regression analysis.
- t-test.
- ANOVA.
- Chi-square test.

14. Suggested Findings Format

If you conduct a survey, the findings can be presented in the following format:

Finding	Interpretation
Most respondents consider AI to be productivity-enhancing	AI supports operational efficiency
Adoption is higher in marketing and customer service	Adoption is faster because of direct customer benefits
Data privacy is a major concern	Strong governance and security controls are necessary
Employees report a need for training	Reskilling is a key requirement for AI implementation

Managers consider AI a decision-support tool	Human judgement remains essential

Important: Do not present the above findings as final research results without conducting an actual survey. They should only be used as expected or proposed findings.

15. Recommendations

1. Organizations should establish clear business objectives before adopting AI.
2. AI implementation should begin with small pilot projects.
3. Continuous AI training and reskilling programs should be provided for employees.
4. Strict privacy policies should be established for customer and employee data.
5. Human review should be mandatory for high-impact decisions.
6. AI systems should be regularly audited for bias and accuracy.
7. Managers should focus not only on technology procurement but also on workflow redesign.
8. Affordable cloud-based AI solutions should be developed for small and medium-sized enterprises.
9. AI-generated information should be verified before being used for financial and legal decisions.
10. Organizations should measure AI performance through productivity, customer satisfaction, revenue, and cost savings.

16. Conclusion

Artificial Intelligence has become an important technology for improving efficiency, personalization, forecasting, and innovation in commerce and management. AI adoption among large enterprises in India is growing, and initiatives such as the Government of India's IndiaAI Mission are supporting the development of the AI ecosystem.

Although AI can increase productivity and create new business opportunities, risks such as privacy concerns, cybersecurity threats, algorithmic bias, job displacement, and skill gaps also exist. Therefore, successful AI adoption requires not only technology but also human judgement, ethical governance, employee training, and data security.

The future of AI is more likely to be about **"Human with Machine" rather than "Human versus Machine."** Organizations that adopt AI not merely as an automation tool but as an integrated system for business strategy, employee development, and customer value creation will be better positioned for success.

17. References

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Important Academic Note: When presenting current data in a research paper, always mention the year, report name, and source alongside every statistic. Market forecasts should be presented as *projected estimates*, not actual facts. Furthermore, never add survey percentages or results without conducting your own primary research.